

Curriculum Vitae

Yue Ma — Experimental Nuclear Physicist

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1. Profile

Experimental nuclear physicist and Senior Research Scientist (permanent) at RIKEN, with twenty years of front-line research and leadership at major international facilities in Japan (J-PARC, SPring-8, KEK) and Germany (GSI, DESY), and a long record of training young researchers and building international collaborations. **Currently experiment spokesperson of J-PARC E73 (hypertriton lifetime), and proposal spokesperson of SPring-8 LEPS2 (nucleon short-range correlations) and the J-PARC antideuteron experiment.**

2. Academic Leadership & Appointments

- J-PARC E73, **hypertriton lifetime — Experiment spokesperson.** Beam time completed; the ${}^4_{\Lambda}\text{H}$ lifetime and ${}^3,4_{\Lambda}\text{H}$ production cross sections are published; the ${}^3_{\Lambda}\text{H}$ lifetime analysis is ongoing.
- SPring-8 LEPS2, **nucleon short-range correlations — Proposal spokesperson.** Probing short-range correlations between nucleons with the Λ hyperon as a probe (ongoing).
- J-PARC antideuteron experiment — **Proposal spokesperson.** Antideuteron–nucleus interaction and annihilation mechanism (ongoing).

3. Signature Achievements

1. **Discovery of the K^-pp ($\bar{K}NN$) kaonic nuclear bound state** — as a founding member, developed the multi-hit TDC and extracted the signal by partial-wave analysis, giving the first evidence for a deeply bound K^-pp state (J-PARC E15). Phys. Lett. B **789** (2019); Phys. Rev. C **102** (2020).
2. **Precise lifetime measurement of the ${}^4_{\Lambda}\text{H}$ hypernucleus** — as spokesperson and corresponding author, proposed the key idea of tagging the fast π^0 with a single high-energy γ and led the measurement, $\tau = 206 \pm 8 \pm 12$ ps (J-PARC E73). Phys. Lett. B **845** (2023); **873** (2026).
3. **First observation of a $\Lambda\pi$ structure near the $\bar{K}N(I=1)$ threshold** — led the Belle data analysis as first author. Phys. Rev. Lett. **130** (2023).
4. **First application of TES microcalorimeters to hadronic-atom spectroscopy** — implemented the SDD data analysis and energy calibration at sub-eV precision (J-PARC E62). Prog. Theor. Exp. Phys. **2016**, 091D01; Phys. Rev. Lett. **128** (2022).
5. **First proposal and demonstration of energy-resolved muon spin spectroscopy (μSR)** — as first and corresponding author, introduced event-by-event positron-energy selection with lead-glass Cherenkov calorimeters and led the demonstration, a $\sim 30\%$ figure-of-merit gain at no cost in beam time (J-PARC MLF). Nucl. Instrum. Meth. A, in press (2026).

4. Education

2006–2009	Doctor of Science , Tohoku University (Sendai)
2004–2006	M.S. in Physical Science, Tohoku University (Sendai)
1998–2002	B.S. in Nuclear Physics , Lanzhou University

5. Appointments

2012–now	Senior Research Scientist (permanent) , RIKEN
2009–2012	Postdoctoral Researcher, Helmholtz Institute Mainz, Germany
2002–2004	Research Trainee, Institute of Modern Physics, CAS

6. Grants and Fellowships

2026–2027	Grant-in-Aid for Transformative Research Areas (A), Publicly Offered Research: search for a Σ -hypertriton ($^3_\Sigma\text{H}$) bound state at J-PARC (within E. Hiyama's research area). PI
2024–2028	Grant-in-Aid for Scientific Research (A): ρ -meson mass modification in nuclei with photon beams. Co-investigator
2023–2025	RIKEN Incentive Research Project: exploring matter–antimatter interactions. PI
2017–2020	JSPS Grant-in-Aid for Young Scientists (A) 17H04842: solving the hypertriton lifetime puzzle. PI

7. Student Supervision

2017–2020	C. Han (PhD), Univ. of Chinese Academy of Sciences
2012–2014	Q. Zhang (PhD), Lanzhou University
2010–2011	D. Lin (PhD); Christina, Pascal (M.S.), Univ. of Mainz

8. Service & Invited Talks

2019–2021	Member, RIKEN Scientists' Assembly Steering Committee; coordinator / chief coordinator, RIKEN Incentive Research Fund (Physics)
2022	Invited talk, 14th Int. Conf. on Hypernuclear & Strange Particle Physics (HYP 2022, Prague)
2025	Invited talk, 15th HYP (HYP 2025, Tokyo)

9. Teaching & Languages

2018/23/24	Invited lecturer, Hadron Physics summer course, Univ. of Chinese Academy of Sciences
Languages	Chinese (native) · English (fluent) · Japanese (fluent)

10. Selected Publications (by leading role and impact; authorship indicated)

More than **120 publications** (including Belle / Belle II collaboration papers, see the final section); the following are first-author / corresponding-author / spokesperson or otherwise leading contributions.

Note: hadron-physics papers are usually listed alphabetically by surname; the **first author** / **corresponding author** / **spokesperson** tags below mark the actual leading role.

1. **Y. Ma et al.**, *First observation of $\Lambda\pi^+$ and $\Lambda\pi^-$ signals near the $\bar{K}N(I=1)$ mass threshold in $\Lambda_c^+ \rightarrow$*

- $\Lambda\pi^+\pi^+\pi^-$ decay, Phys. Rev. Lett. **130**, 151903 (2023) [Belle]. [First author]
2. T. Akaishi et al., Precise lifetime measurement of ${}^4_\Lambda\text{H}$ hypernucleus using in-flight ${}^4\text{He}(K^-, \pi^0)$ reaction, Phys. Lett. B **845**, 138128 (2023). [Spokesperson & corresponding author]
 3. T. Akaishi et al., Measurement of ${}^3,4\text{He}(K^-, \pi^0){}_\Lambda^3,4\text{H}$ reaction cross section and evaluation of hypertriton Λ binding energy, Phys. Lett. B **873**, 140163 (2026). [Spokesperson]
 4. Y. Ma et al., Demonstration of energy-resolved muon spin spectroscopy using a Cherenkov calorimeter, Nucl. Instrum. Meth. A, in press (2026). [First author & corresponding author]
 5. Y. Ma et al., γ -ray spectroscopy study of ${}^{11}_\Lambda\text{B}$ and ${}^{12}_\Lambda\text{C}$, Eur. Phys. J. A **33**, 243 (2007). [First author]
 6. S. Ajimura et al., K^-pp , a $\bar{K}$ -meson nuclear bound state, observed in ${}^3\text{He}(K^-, \Lambda p)n$ reactions, Phys. Lett. B **789**, 620 (2019). [Major author]
 7. T. Yamaga et al., Observation of a $\bar{K}NN$ bound state in the ${}^3\text{He}(K^-, \Lambda p)n$ reaction, Phys. Rev. C **102**, 044002 (2020). [Major contributor]
 8. T. Hashimoto et al., Strong-interaction effects in kaonic-helium isotopes at sub-eV precision with X-ray microcalorimeters, Phys. Rev. Lett. **128**, 112503 (2022). [Major contributor]
 9. B. S. Henderson et al., Hard two-photon contribution to elastic lepton-proton scattering determined by the OLYMPUS experiment, Phys. Rev. Lett. **118**, 092501 (2017). [Major contributor; led the PbF₂ luminosity monitor]
 10. K. Hosomi et al., Precise determination of ${}^{12}_\Lambda\text{C}$ level structure by γ -ray spectroscopy, Prog. Theor. Exp. Phys. **2015**, 081D01 (2015). [Second author]

11. Belle / Belle II Collaboration Papers

As a member of the Belle / Belle II collaboration since 2023, co-author of 70+ collaboration papers; representative papers by theme:

I. B -meson CP violation & CKM matrix elements Time-dependent CP violation ($\sin 2\phi_1$, etc.), $|V_{cb}|/|V_{ub}|$, and the CKM angle ϕ_3 .

1. Measurement of $\sin 2\phi_1$ in $B^0 \rightarrow J/\psi K_S^0$ decays with a graph-neural-network flavor tagger, Phys. Rev. D **110**, 012001 (2024) [Belle II].
2. Measurement of CP violation in $B^0 \rightarrow K_S^0 \pi^0$ decays, Phys. Rev. Lett. **131**, 111803 (2023) [Belle II].
3. Determination of $|V_{cb}|$ using $\bar{B}^0 \rightarrow D^{*+} \ell^- \bar{\nu}_\ell$ decays, Phys. Rev. D **108**, 092013 (2023) [Belle II].
4. Measurement of inclusive $B \rightarrow X_u \ell \nu$ partial branching fractions and $|V_{ub}|$, Phys. Rev. D **113**, 032004 (2026) [Belle II].
5. Determination of the CKM angle ϕ_3 from a combination of Belle and Belle II results, J. High Energy Phys. **10** (2024) 143 [Belle/Belle II].

II. Rare decays & lepton-flavour universality First evidence for $B^+ \rightarrow K^+ \nu \bar{\nu}$, $R(D^*)$ and other LFU tests, and τ lepton-flavour-violation searches.

1. Evidence for $B^+ \rightarrow K^+ \nu \bar{\nu}$ decays, Phys. Rev. D **109**, 112006 (2024) [Belle II].
2. Test of light-lepton universality with a measurement of $R(D^*)$ using hadronic B tagging, Phys. Rev. D **110**, 072020 (2024) [Belle II].
3. First measurement of $R(X_{\tau/\ell})$ as an inclusive test of the $b \rightarrow c\tau\nu$ anomaly, Phys. Rev. Lett. **132**, 211804 (2024) [Belle II].
4. Tests of light-lepton universality in angular asymmetries of $B^0 \rightarrow D^{*-} \ell \nu$ decays, Phys. Rev. Lett. **131**, 181801 (2023) [Belle II].
5. Measurement of the $B^+ \rightarrow \tau^+ \nu_\tau$ branching fraction with a hadronic tagging method, Phys. Rev. D **112**, 072002 (2025) [Belle II].
6. First search for $B \rightarrow X_s \nu \bar{\nu}$ decays, Phys. Rev. Lett. **136**, 231801 (2026) [Belle II].
7. Search for lepton-flavor-violating τ decays to $\ell\alpha$, J. High Energy Phys. **08** (2025) 155 [Belle].
8. Search for a $\tau^+\tau^-$ resonance in $e^+e^- \rightarrow \mu^+\mu^-\tau^+\tau^-$ events, Phys. Rev. Lett. **131**, 121802 (2023) [Belle II].

III. Charm/strange baryon spectroscopy & exotic hadrons(closely aligned with my own research) Ξ_c decays and CP, the first observation of $D_{s0}^*(2317)$ radiative decay, P_{cs} pentaquarks, and $\Xi^0 p/\Omega^- p/\Omega^- n$ dibaryon searches.

1. *Observation of the radiative decay $D_{s0}^*(2317)^+ \rightarrow D_s^{*+}\gamma$* , Phys. Rev. Lett. **136**, 241901 (2026) [Belle/Belle II].
2. *Search for $P_{cs}(4459)$ and $P_{cs}(4338)$ in $\Upsilon(1S, 2S)$ inclusive decays*, Phys. Rev. Lett. **135**, 041901 (2025) [Belle/Belle II].
3. *Search for CP violation in $\Xi_c^+ \rightarrow \Sigma^+ h^+ h^-$ and $\Lambda_c^+ \rightarrow p h^+ h^-$ decays*, Phys. Rev. D **113**, 032017 (2026) [Belle II].
4. *Observation of the singly Cabibbo-suppressed decays $\Xi_c^+ \rightarrow p K_S^0, \Lambda \pi^+, \Sigma^0 \pi^+$* , J. High Energy Phys. **03** (2025) 061 [Belle/Belle II].
5. *Measurement of branching fractions of $\Xi_c^0 \rightarrow \Xi^0 \pi^0, \Xi^0 \eta, \Xi^0 \eta'$* , J. High Energy Phys. **10** (2024) 045 [Belle/Belle II].
6. *Observation of $B^+ \rightarrow \Sigma_c(2455)^{++} \Xi_c^-$ and $B^0 \rightarrow \Sigma_c(2455)^0 \Xi_c^-$* , Phys. Rev. D **112**, L051101 (2025) [Belle/Belle II].